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PALAEOMAGNETIC CORRELATIONS OF PRECAMBRIAN FORMATIONS OF EAST-CENTRAL AFRICA AND THEIR TECTONIC IMPLICATIONS

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ABSTRACT

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Palaeomagnetic poles derived from Precambrian formations can be valuable for determining relative, and sometimes absolute, ages of the formations. In this paper palaeomagnetic results are presented from a variety of these formations in Tanzania and Zambia. The Ikorongo Group sediments of Tanzania give a pole at 80°E,25°S commensurate with an age of 900—1000 m.y. The lower Buanji Series of southern Tanzania yields a pole at 263°E,87°N indicating an age of either Late Precambrian (c. 650 m.y.) or Early Cambrian. The Plateau Series outcrop at the southern end of Lake Tanganyika gives several poles falling on the Late Precambrian to Ordovician apparent polar wander loop recognized by McElhinny et al. (1974), and a small amount of evidence from the Abercorn Sandstone and southern part of the Plateau Series outcrop suggests an age of c. 900 m.y. for these rocks. Dating of formations at the southern end of the Lake Tanganyika depression gives an estimate of 1500 m for the minimum amount of downthrow at this end of the rift system. Five sites from the Mbozi gabbro—syenite complex of southern Tanzania give a pole at 68°E,72°N and two sites from Mbala dolerites of Zambia yield a pole close to one from the Bukoban dolerites of Tanzania and a similar age (c. 806 m.y.) is suggested.

Some palaeomagnetic information is now available from all the Proterozoic platform sediments margining the Tanganyika craton and a correlation scheme is given which incorporates this information together with geochronological data. These formations postdate geosynclinal sequences involved in the Kibaran (c. 1300 m.y.) and Irumide (c. 1100 m.y.) mobile belts, and geological environment and situation demonstrate that the Tanganyika craton was subject to intermittent uplift between about 1000 m.y. and Cambrian times.

INTRODUCTION

The cratonic regions of east-central Africa are partially overlain by Precambrian sedimentary formations (Fig. 1) predominantly of clastic lithology, and which in several cases are partially incorporated into later mobile belts. Their maximum ages are loosely defined by age dates derived from basement rocks on which they lie, and minimum ages are sometimes available from igneous rocks intrusive into them. A summary of known age relationships of

Precambrian formations in central Africa based on information available to 1970 is given by Cahen (1970).

It has already been suggested (McElhinny et al., 1968) that palaeomagnetism may provide a means of dating Precambrian formations whose ages cannot closely be defined by other means. This paper presents palaeomagnetic results from Precambrian formations of northern Zambia and several parts of Tanzania which complement results already obtained from the Late Precambrian Bukoban System (Piper, 1972) so that some palaeomagnetic information now exists for all the largely undeformed platform sediments cropping out between northeast Zambia and western Kenya. These results are summarized and implications to the historical geology of the region examined. In addition palaeomagnetic results are reported from the Late Precambrian Mbozi Complex of Tanzania and the Mbala dolerites of the Zambia—Tanziania border region which help to improve definition of the Precambrian apparent polar-wander curve, and in the latter case provide some control on the age of intruded sediments.

FRAMEWORK OF BASEMENT ROCKS

The Precambrian platform sediments are deposited unconformably on a basement comprising a sequence of superimposed mobile belts between cratonic nuclei (Clifford, 1968 and Fig. 1, inset). These predominantly granitic nuclei comprise the Tanganyika craton, for which geochronological studies yield rather uniform ages around 2600 m.y., the Bangweulu craton (> 1800 m.y.) in the south and the Kasai craton (> 2865 m.y.) in the east. The Tanganyika craton is separated from the other two by the NW trending Ubendian—Rusizian belt with post-orogenic ages in the range 1700—1800 m.y. (Cahen and Snelling, 1966), which is in turn cut by the NNE trending Kibaran belt with synorogenic ages of c. 1300 m.y. and the sub-parallel Irumide belt with synorogenic episodes c. 1100 m.y. (Ray, 1970). The Irumide trend extends from Rhodesia to Tanzania (Fig. 1, inset, Vail et al., 1968) where it probably reaches at least as far as Iringa (Ackermann, 1960) and ages of c. 900 m.y. occur in the later Mozambique belt in line with this trend (Cahen and Snelling, 1966). Both the Irumide and to a lesser extent the Kibaran belts have been partially incorporated in the "Pan African" Lufilian—Mozambique mobile belts. The history of these latter belts (Clifford, 1967) is proving to be more complicated than previously recognized (L. Cahen, personal communication, 1973) and early phases of the Lufilian/Katangan orogeny are now known to have occurred before 700 m.y. with late-stage thermal episodes occurring in the Early Palaeozoic.

The overprinting of successively younger orogenic cycles on the stratigraphy has made the task of recognizing the correlating formations from place to place a difficult one, and only in a few areas has it so far proved possible to correlate platform sediments on the cratons with those involved in the mobile belts. Before radiometric information became widely available, correlations necessarily had to be made on lithological similarities and were widely discussed in the literature of a decade or more ago (e.g. Harpum, 1955).

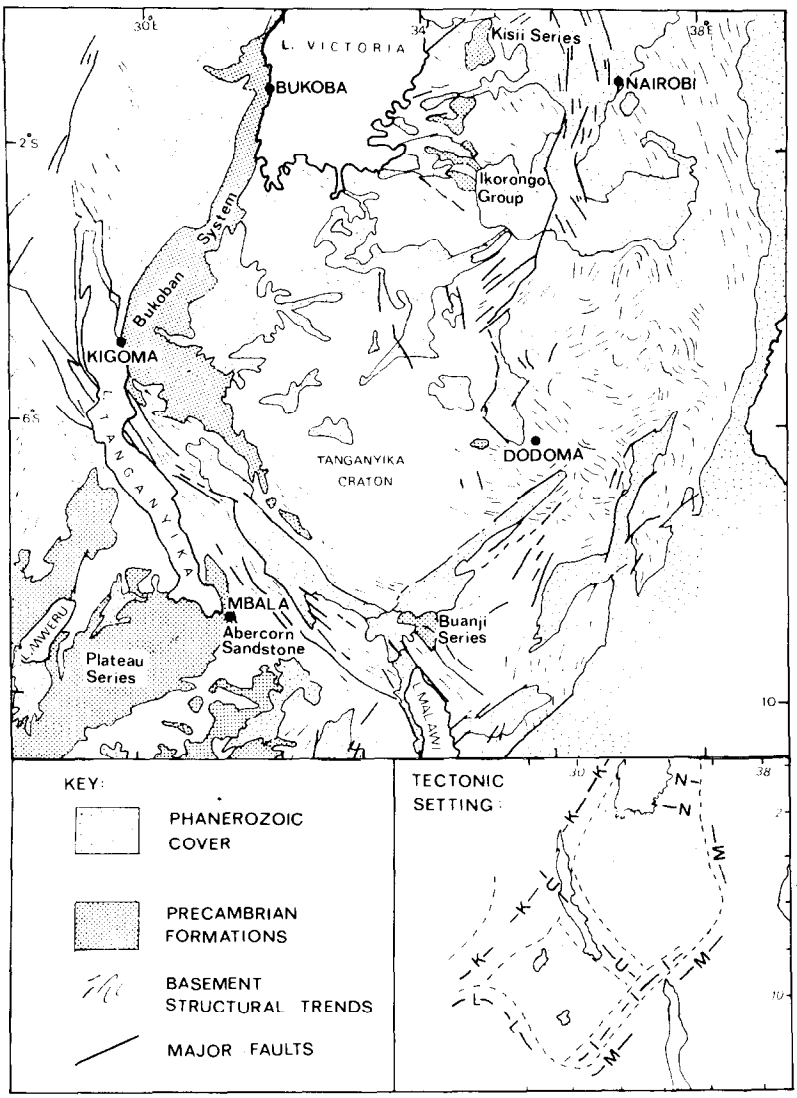


Fig. 1. Distribution of largely undeformed Proterozoic platform sediments. Structural trends in folded and metamorphosed basement rocks are compiled from Harpum (1955) and Hepworth (1972) for the areas between latitudes 3.5° and 7.5°S, and longitudes 35° and 38°E, P.W.G. Tanner (personal communication, 1973) and the Carte Tectonique de l'Afrique. The inset diagram shows the distribution of orogenic belts in relation to the cratonic areas (stippled). The belts are: N, Nyanzian (> 2200 m.y.); U, Ubendian (> 1800 m.y.); K, Kibaran (c. 1300 m.y.); I, Irumide (c. 1100 m.y.); L, Lufilian and M, Mozambique (750—450 m.y.).

GEOLOGICAL SETTING OF PRECAMBRIAN PLATFORM SEDIMENTS

In the area under consideration Precambrian platform sediments occupy two large outcrops, the Bukoban System, the Plateau Series and several small outcrops around the margins of the Tanganyika craton.

The Bukoban System appears to have been laid down in subsiding basins of roughly similar extent (Halligan, 1962) to its present arcuate outcrop around the western side of the Tanganyika craton. It consists predominantly of argillaceous and arenaceous sediments with some thin limestones and a major volcanic sequence (Gagwe Amygdaloidal Lavas) near the top. Members of this system give virtual geomagnetic poles (VGPs) which fall on an arc of about 80° (Piper, 1972), their location on the arc corresponding generally to stratigraphic position. The latter part of the arc is dated by ages on the Gagwe Amygdaloidal Lavas (813 ± 30 m.y.) and dolerite intrusions in the Bukoban Sandstone Formation (806 ± 30 m.y.). By correlation with the Musindozi Formation of Burundi, the pole for the Kigonero Flags near the centre of the arc is dated at > 890 m.y. and the Burundi extension of the Bukoba Sandstone Formation is dated at > 995—1000 m.y. (N.J. Snelling, unpublished result, L. Cahen, personal communication, 1973).

Precambrian sedimentary formations crop out over much of the northern and Luapula Provinces of Zambia where they comprise medium to course-grained sandstones with subordinate siltstones, shales and conglomerates. These are largely undeformed, but become increasingly folded along NE lines towards the Mansa area (29°E,10°S) in the west of the region. In the northeast of the outcrop Page (1960) distinguishes a Lower Plateau Series as a molasse deposit intruded by the Kate Granite (> 1780 m.y., Cahen and Snelling, 1966), a post-orogenic granite to the Ubendian belt and cropping out over an area in excess of 250 km^2 straddling the Zambia—Tanzania border. This is separated by an unconformity from an Upper Plateau Series which Cahen (1970) interprets as a foreland equivalent of deposits involved in the Irumide belt (c. 1100 m.y.) to the west. That much of the so-called Plateau Series is a molasse deposit to the Ubendian belt is shown by mapping in the western part of the outcrop area (Thieme, 1970, 1971) where, although separated from porphyry and syntectonic granite basement by an unconformity, the Series is folded into these rocks along similar lines to earlier basement folding. An earlier suggestion (Reeve, 1963) that the Plateau Series may be separated into earlier strongly folded and later weakly folded units has not, however, been confirmed by this mapping and the only current minimum age control on the Plateau Series of this general area is given by a date of 1390 ± 70 m.y. on the Lusenga Syenite intrusion at 29.2°E,9.3°S (Thieme, 1971). Other sediments of the same region, the Luapula Beds, contain Lower Palaeozoic microfossils (Anon, 1972, note 116) and are thus much younger in age; the term "Plateau Series" is a loose one, and embraces sediments of rather uniform lithology (mostly sandstones and siltstones) with an overall age range in excess of 1000 m.y.

Sampling of the Plateau Series for the present study has been concentrated in tectonic blocks downfaulted into the Lake Tanganyika depression. It is not yet certain whether these sediments are merely preserved here in the downfaulted trough, or whether they merit comparison with other parts of the Plateau Series outcrop. In view of the slender evidence that most of the out-

crop is much older in age they might better be distinguished as a separate group and L. Cahen (personal communication, 1973) has suggested the term "Downfaulted Plateau Series".

Overlying the Upper Plateau Series of Page (1960), unconformably in the Lake Tanganyika region, is the Abercorn Sandstone Formation, a minimum age for which is given by two mineral ages of 940 ± 40 m.y. (pyroxene) and 678 ± 30 m.y. (plagioclase) on a dolerite from Kasanga in Tanzania (Vail et al., 1968). This dolerite is one of a suite of intrusions into the Abercorn Sandstone referred to here as the Mbala dolerites, and which have also been sampled for palaeomagnetic study. The sandstone comprises about 1000 m of medium orthoquartzite in the type area, with current bedding directions implying derivation from the west and northwest where the large Kate Granite outcrop (> 1740 m.y.) appears to have been a likely source region. Extent of the Formation is unknown but deposits near Mpika (31.6°E,12°S) known as the Bemba Beds and regarded as post-orogenic to the Irumide belt (Ackermann and Forster, 1960) may correlate with it (Cahen, 1970).

PALAEOMAGNETIC RESULTS

Sampling and laboratory treatment

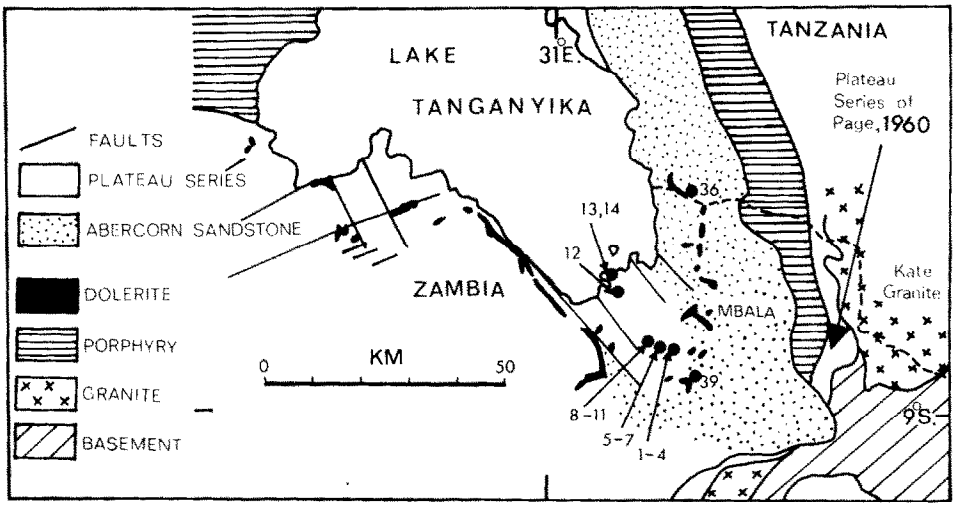
The bulk of the collections were sampled using a portable coring drill (Doell and Cox, 1967) although material from the Mbala dolerites, Abercorn Sandstone and Ikorongo Group were lump sampled (between four and seven samples being collected from each site) and cored in the laboratory. All samples were oriented while still attached to the outcrop by sun and/or magnetic bearings.

Cores were cut into 1-inch long cylinders and total NRMs measured using a PAR SM-2D spinner magnetometer. Two samples were selected from each igneous site and one from each sediment site and subjected to stepwise a.f. demagnetization; steps of 100 or 200 Oersteds (Oe) to fields between 600 and 1600 Oe were chosen, and a stability index (Briden, 1972) computed for each step. From demagnetization characteristics and maximum stability indices, one or more optimum fields were chosen from the behaviour of the pilots for treatment of the remaining samples from the site.

Plateau Series, Zambia

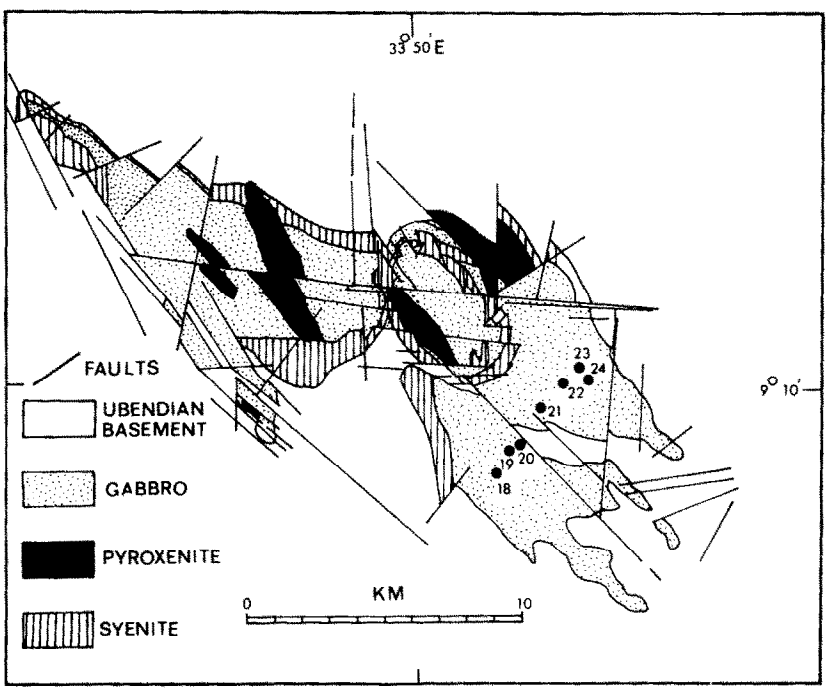
This series was sampled at three areas between Mbala and Mpulungu in red siltstones (Fig. 2A), and the sites at these localities in order of increasing age are: sites 1—4, sites 5 and 6, sites 7—10, and site 11. Three sites (12—14) were also sampled from red and purple sandstones and siltstones at Mpulungu on the shore of Lake Tanganyika, and three further sites (15—17) were taken from red sandstones 280 km south of Mbala, where they dip 40° to the north.

Samples have intensities mostly varying between $0.2 \cdot 10^{-4}$ and $0.6 \cdot 10^{-5}$ Gauss (G) with a log mean of $0.9 \cdot 10^{-5}$ G; sites 6 and 15 are significantly



A

Fig. 2A. Geological map of the area around the southern end of Lake Tanganyika with palaeomagnetic sampling sites.



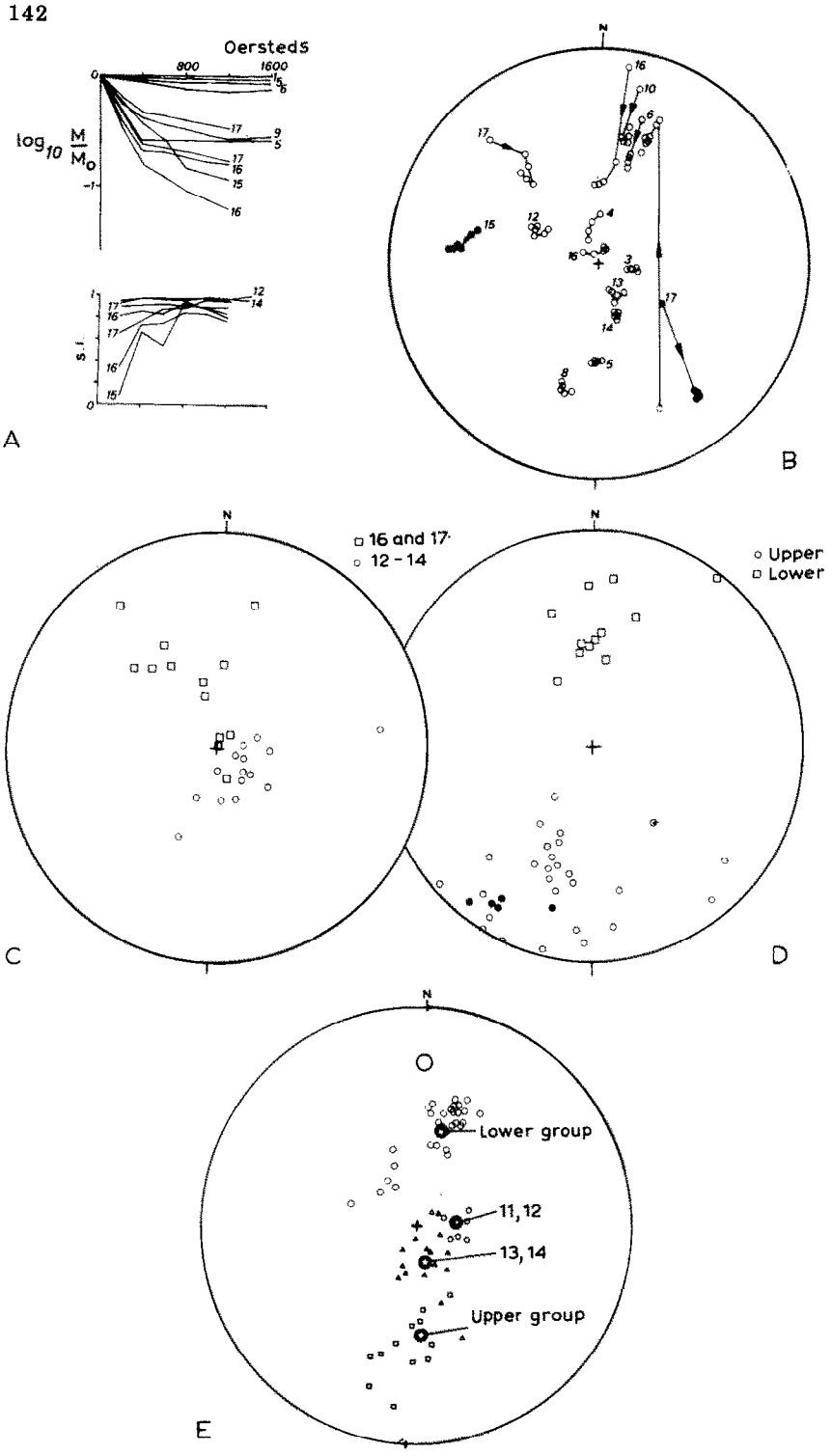
B

Fig. 2B. Geological map of the Mbozi complex after Brock (1968) with palaeomagnetic sampling localities.

weaker ($0.2 - 0.5 \cdot 10^{-5}$ G) and site 5 is stronger ($0.3 \cdot 10^{-3}$—$0.4 \cdot 10^{-4}$ G) than the remainder of the collection and appears to have acquired a large secondary magnetization possibly due to lightening. Total NRMs are moderate to well grouped for all sites except 3, 5, 9, 15 and 16. Most pilot samples showed slight to moderate changes in direction with stepwise demagnetization (Fig. 3A) and characteristically lose less than 10% of their moments after demagnetizing to 1600 Oe (Fig. 3B). The exceptions to this rule are the pilots from sites 5 and 9 which lose between 50 and 80% of their moments in demagnetizing to 1000 Oe and show marked direction changes with the first 200-Oe treatment. After cleaning at 1000 Oe the Fisher precision parameter k of site 9 increases and it acquires a direction close to that of sites 3, 4, 7 and 10. Site 5 which is random (Irving, 1964, table 4.3) before cleaning becomes statistically significant after cleaning and acquires a direction closely antiparallel to that of sites 1, 2 and 8 but different from that of sites 3, 4, 7, 9 and 10. All pilot samples from sites 1—10 have optimum stability indices greater than 0.9 which they reach after demagnetizing in fields between 800 and 1200 Oe and the remaining samples from each site have been cleaned in these fields. With this treatment all sites except site 2 (which becomes random) show increase in precision. The tendency of the sites 1—10 to fall into two groups is enhanced by this treatment, and the combined direction of the first group (1, 2 and 8 before cleaning and 1, 5 and 8 after cleaning) moves from $D = 179°$, $I = 29°$ ($k = 18$) to $D = 177°$, $I = -37°$ ($k = 99$). The direction of the second group (significant sites 3, 4, 6, 7 and 10 before cleaning, significant sites 3, 4, 6, 7, 9 and 10 after cleaning) moves from $D = 17°$, $I = -21°$ ($k = 18$) to $D = 13°$, $I = -41°$ ($k = 19$). The latter direction is close to, but significantly removed from the present geomagnetic field direction in this region, and presence of a reversal of magnetization in the former group argues in favour of primary remanence. The derived virtual geomagnetic pole positions are 205°E,60°S for the former group and 173°E,71°N for the latter group.

The former group includes the uppermost sites at the three localities where more than one site was sampled and the latter group includes the lower sites. These three localities are in fine to medium grained siltstones dipping at a few degrees to the east, and the total thickness sampled is not more than about 10 m. The sections may overlap one another although exposure is not good enough to be certain of this. Difference in the derived pole between the former and the latter groups of sites probably represents a real polar shift but it is currently not possible to be sure of this.

The remaining sites have similar magnetic properties to sites 1—10; site 11 becomes significant with a.f. cleaning and the remaining sites show an improvement in precision. The directions of magnetization are however quite different from sites 1—10. Site 11 whose stratigraphic position is uncertain has a direction close to site 12 and near $D = 85°$, $I = 68°$, and sites 13 and 14 which are at a similar, probably slightly higher stratigraphic horizon to site 12, have southerly negative directions of magnetization and combine to give a mean direction $D = 159°$, $I = 72°$ (Fig. 3E, Table I).


Oersteds
800
1600
$\log_{10} \frac{M}{M_0}$
S.I.
A
N
B
C
16 and 17
12 - 14
D
Upper
Lower
E
Lower group
11, 12
13, 14
Upper group

The generally more weakly magnetized ($0.2 \cdot 10^{-4}$—$0.4 \cdot 10^{-5}$ G) samples from the three sites (15, 16 and 17) of the southern outcrop show a systematic fall in their moments on demagnetization with loss of 50—90% of their total moments after demagnetization to 1000 Oe (Fig. 3A). Directions of magnetization also change with a tendency for largest movements in the range 0—400 Oe (Fig. 3B). Stability indices reach highest values at 800 Oe (Fig. 3B) and all the samples from these sites have accordingly been demagnetized at this field. After demagnetization one site (16) random at the total NRM level becomes significant and the single site (17) which was significant before cleaning and has an intensity an order of magnitude higher than the other two sites slightly improves its precision. The two directions from these cleaned sites (D = 16°, I = —72° and D = 325°, I = —37°) are however quite different.

Abercorn Sandstone

In this paper this term is restricted to the lithologically distinct formation of the type area although Page (1960) includes the Plateau Series of sites 1—14 as "Upper" Abercorn Sandstone. Five samples collected 16 km west of Mbala combine (Fig. 5, Table I) to give a mean direction of $D = 319°, I = -14°$ (k = 26). The samples have been treated by a.f. demagnetization up to 1600 Oe but show no significant change in magnitude or direction of their magnetic vectors, suggesting that the entire remanence is carried by a member of the hematite—ilmenite series (Nagata, 1967).

Mbozi complex, Tanzania

The Mbozi complex is a composite intrusive body composed predominantly of gabbro and syenite. It is 32 km long and 13 km wide, intrudes metasediments of the Ubendian belt, and is elongated parallel to this belt and the Tunduma section of the East African Rift System. Mapping by Brock (1968) has shown that the oldest and predominant rocks of the complex are layered gabbros which were deformed, faulted and intruded by dolerite dykes before intrusion and metasomatism by alkaline rocks.

The seven sites 18—24 form an approximate traverse across the body from

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Fig. 3. A. Demagnetization characteristics and behaviour of stability indices (S.I.) with progressive demagnetization for pilot samples of the Plateau Series, Zambia. B. Directional behaviour of pilot samples with progressive demagnetization in steps of 200 or 400 Oe. C. A.f. cleaned sample directions for sites 16 and 17 (squares), and uncleaned directions for sites 12 to 14 (circles). D. Uncleaned sample directions for lower sites (3, 4, 6, 7, 9 and 10, squares) and upper sites (1, 2, 5 and 8, circles) from the Plateau Series. E. A.f. cleaned sample directions for all sites from the Lake Tanganyika outcrop of the Plateau Series. The stars are mean directions for each group of sites, and directions of samples from site 5 have been reversed in D and E for comparison with the remainder of group. Throughout this paper open symbols refer to upper and closed symbols to lower hemisphere plots. Large circle is present earth's field direction.

TABLE I

Plateau Series, Zambia

Site No.	N	D	I	R	k	α_{95}	Virtual geomagnetic pole positions	
							°E	°N
Lake Tanganyika outcrop, higher sites (31°E,9°S)								
1	4	171	−44	3.9	55.0	12	197	−54
2	3	177	−13	2.6	4.5	66	198	−74
5	3	—	—	—	—	—	—	—
8	5	195	−19	4.8	18.2	18	249	−66
Lake Tanganyika outcrop, lower sites (31°E,9°S)								
3	4	—	—	—	—	—	—	—
4	5	1	−36	3.8	3.3	49	208	79
6	5	10	−16	4.7	13.0	22	118	80
7	5	23	−25	4.7	13.5	22	113	67
9	6	—	—	—	—	—	—	—
10	8	30	5	7.9	98.1	6	101	58
Lake Tanganyika outcrop, Mpulungu (31°E,9°S)								
11	5	—	—	—	—	—	—	—
12	4	91	−62	3.6	7.2	37	164	6
13	5	123	−70	4.9	41.4	12	181	−11
14	6	175	−65	5.7	19.9	15	206	−33
Southern outcrop (31°E, 11°S)								
16	6	—	—	—	—	—	—	—
17	7	327	−18	6.4	11.0	19	298	57
Miscellaneous results, Zambia								
Mbala Dolerites (31.5°E,9.0°S)								
36	—	—	—	—	—	—	—	—
39	11	119	23	6.8	2.4	38	103	−26
Abercorn Sandstone (31.5°E,9.0°S)								
45	5	318.5	−14.4	4.8 ($d\varphi$ = 8.1, $d\chi$ = 15.8)	25.7	15.4	300.	49

N is the number of samples or sites as indicated; R is the resultant of N unit vectors; D = declination and I = inclination of remanent magnetism; $k = (N-1)/(N-R)$ = estimate of Fisher's (1953) κ parameter; α_{95} = semi angle of cone of 95% confidence for mean directions.

TABLE I (continued)

Cleaning field (Oe)	N	D	I	R	k	α_{95}	Virtual geomagnetic pole positions	
							°E	°N
900	5	172	−43	4.9	40.7	12	199	−55
900	—	—	—	—	—	—	—	—
1000	3	353	+38	2.9	25.9	24	197	−59
1000	5	185	−30	4.9	52.1	11	221	−64
							paleomagnetic pole	
all sites	3	176.8	−37.2	3.0	89.8	13.1	205	−60
		($d\varphi$ = 9.0, $d\chi$ = 15.4)						
1000	4	32	−35	3.5	5.7	42	142	57
1000	6	322	−59	5.9	43.4	10	252	44
800	5	15	−39	4.9	55.3	10	164	71
1000	5	16	−42	5.0	90.3	8	167	69
1000	6	19	−30	6.0	188.3	5	145	71
1000	5	9	−33	5.0	218.1	5	166	77
							palaeomagnetic pole	
all sites	6	13	−41.0	5.7	18.7	15	173	71
		($d\varphi$ = 11.7, $d\chi$ = 19.3)						
1000	5	90	−70	5.0	85.7	8	174	7
1000	3	81	−67	3.0	121.7	11	170	12
1000	8	151	−64	7.7	23.1	12	188	−30
1000	7	178	−80	6.7	23.9	13	210	−11
							palaeomagnetic pole	
sites 11 and 12	2	85.4	−68.4	2.0	696.7	10	172	10
		($d\varphi$ = 13.5, $d\chi$ = 16.1)						
sites 13 and 14	2	159.3	−71.9	2.0	4.2	39.5		
		($d\varphi$ = 61.2, $d\chi$ = 69.5)						
800	6	16	−72	5.3	7.1	27	200	40
800	7	325	−37	6.6	14.6	16	278	54
300	7	87	36	5.6	4.3	33	98	13
600	9	80	43	7.7	6.2	22	99	10
mean of two sites		84	40	2.0	19.3	18	100	9
				($d\varphi$ = 12.9, $d\chi$ = 21.6)				

south to north (Fig. 2B) and are confined to the gabbro facies. On demagnetization intensities fall steeply and nearly exponentially (Fig. 4A) from initial values in the range $0.2 \cdot 10^{-2}$—$0.2 \cdot 10^{-3}$ G with, as a rule, only a few percent of their moments remaining after treatment in 500 Oe.

There is no tendency for the vectors to reach a stable end-point and directions show generally poor stability with a.f. treatment; pilot samples from sites 19, 20 and 21 demonstrate erratic behaviour during initial demagnetization and other pilots show little tendency to reach a stable end-point although 18 and 23 remain moderately stable throughout the range of the treatment (Fig. 4B). The continuous fall in moment with demagnetization results in no well-defined optimum stability index (Fig. 4A) and for all sites it was found necessary to stepwise demagnetize more than one sample; the best cleaning field was then chosen as that giving the best within-site grouping. Four sites (18, 21, 23 and 24) are significant at the total NRM level and three have northerly declinations and shallow inclinations. All sites become significant

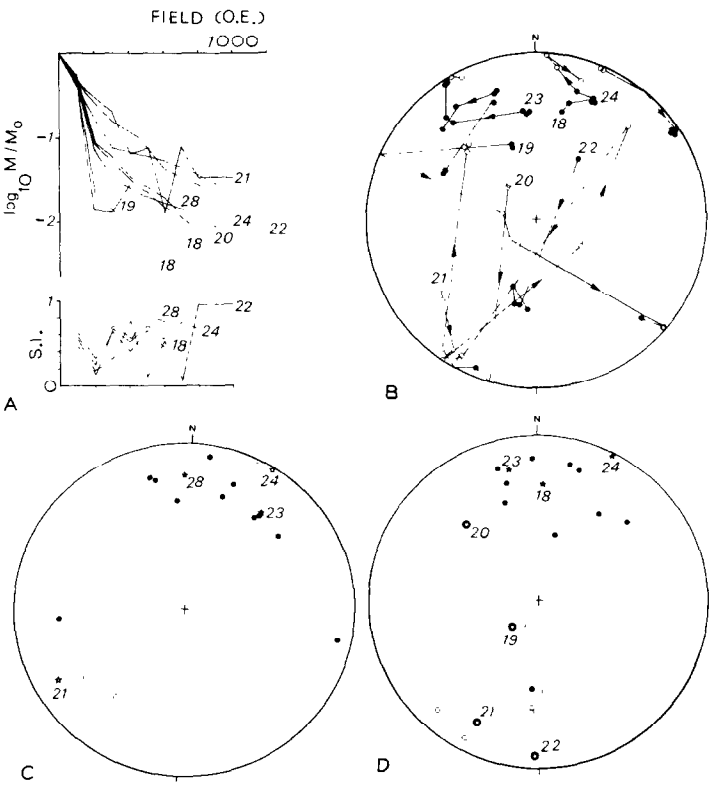


Fig. 4. A. Behaviour of magnetic moments and stability indices for pilot samples from the Mbozi complex, Tanzania, with progressive a.f. demagnetization in steps of 100 Oe. B. Directional behaviour of some pilot samples. C. Sample and site mean directions before cleaning. D. After cleaning. Stars are site mean directions.

TABLE II

Mbozi Complex, Tanzania (33°E,9°S)

Site No.	Site mean statistics before cleaning						Cleaning field (Oe)	Site mean statistics after cleaning						Virtual geomagnetic pole positions	
	N	*D*	*I*	*R*	*k*	α_{95}		*N*	*D*	*I*	*R*	*k*	α_{95}	°E	°N
18	4	357	13	3.9	22.9	20	400	4	2	21	3.8	12.7	27	40	70
19	3	—	—	—	—	—	300	3	222	−65	2.8	13.1	36	242	−23
20	3	—	—	—	—	—	300	3	317	−27	3.0	56.3	16	292	48
21	3	238	−10	2.7	5.9	56	400	4	207	−13	3.8	12.9	27	273	−59
22	3						600	3	184	−6	2.6	5.6	58	230	−78
23	4	35	19	3.2	3.6	56	300	4	348	14	3.8	14.1	25	356	70
24	4	29	−1	3.4	5.1	45	200	4	29	0	3.4	4.8	47	107	60
	4	30.0	11.5	3.7	10.3	30		5	10.1	11.2	4.8	18.5	18	Paleomagnetic pole	
														67.7	72.2
														($d\varphi$ = 9.4, $d\chi$ = 18.5)	

Palaeomagnetic pole position: 97.7°E,56.7°N
($d\varphi$ =15.5, $d\chi$ = 30.6)

Mean obtained by reversing site 21

Mean obtained by excluding sites 19 and 20, and reversing sites 21 and 22

after cleaning and group mean directions of sites 18, 23 and 24 remain substantially the same although site 21 attains a direction more nearly antiparallel to these other sites and is close to the cleaned direction of site 22. The end result is that two sites (21 and 22) are reversed with respect to three of the remainder (18, 23 and 24), and when these are reversed the combined five sites give an overall mean of $D = 10.1°$, $I = 11.2°$ (Table II, Figs. 4C and D) with a palaeomagnetic pole at 68°E,72°N. This pole is significantly removed from the present pole and presence of an exact reversal within the complex is evidence that the remanence is primary.

Sites 19 and 20 are anomalous with respect to the remainder, and since they occur between sites of opposite polarity might possibly have acquired their magnetism during a polarity transition. However, site 19 has intensities (*Jo*) in the range $0.4 \cdot 10^{-2}$ and $0.1 \cdot 10^{-3}$ G and Königsberger ratios (*Qo*) between 0.04 and 0.6, and site 20 has *Jo*-values ranging from $0.4 \cdot 10^{-1}$ to $0.2 \cdot 10^{-3}$ G, and *Qo*-ratios from 0.5 to 15; this compares with the sites 21 and 22 ($Jo = 0.7 \cdot 10^{-2}$—$0.1 \cdot 10^{-3}$ G, $Qo = 1$—9) and the sites 18, 23 and 24 of opposite polarity ($Jo = 0.6 \cdot 10^{-2}$—$0.2 \cdot 10^{-3}$ G, $Qo = 0.6$—12). The intensities, susceptibilities and demagnetization characteristics of the two anomalous sites are comparable to the remaining sites, and give no reason for suspecting that they acquired their magnetization during a reversal. Their directions may merit a tectonic interpretation since the gabbros are visibly deformed in places and these sites may have acquired their magnetism prior to deformation. In addition the complex has been subjected to several periods of faulting including at least two associated with the Miocene rift faulting (Brock, 1968); in this context it is noted that the two anomalous sites lie close to a major fault cutting the complex (Fig. 2A).

Mbala Dolerites, Zambia—Tanzania border

The Abercorn Sandstone formation is extensively intruded by dolerite along the southeastern margin of Lake Tanganyika (Fig. 2A). Most of these intrusions are highly weathered and visible on the surface only as loose boulders. Of the two significant sites of this collection, site 36 is from a sill outcropping at Kalambo Falls on the Zambia—Tanzania border and site 39 is from an intrusion in the Lunzua valley south of Lake Tanganyika. These dolerites contain fresh plagioclase feldspar with pyroxene and titanomagnetite in an intersertal relationship. The pyroxene is uralitized to a variable degree and the titanomagnetite grains are marginally altered; they show a minor amount of subdivision by ilmenite lamellae. Intensities of magnetization are high ($8 \cdot 10^{-1}$—$2 \cdot 10^{-2}$ G) and only site 39 is significant before cleaning. With a.f. treatment magnetic moments of the samples with high initial values drop dramatically and are accompanied by large changes in direction (Fig. 5A). The highest stability indices are reached where the demagnetization curves begin to flatten out about 600—700 Oe when around 1% of the total moment remains (Fig. 5A). After cleaning sites 36 and 39 are significant and

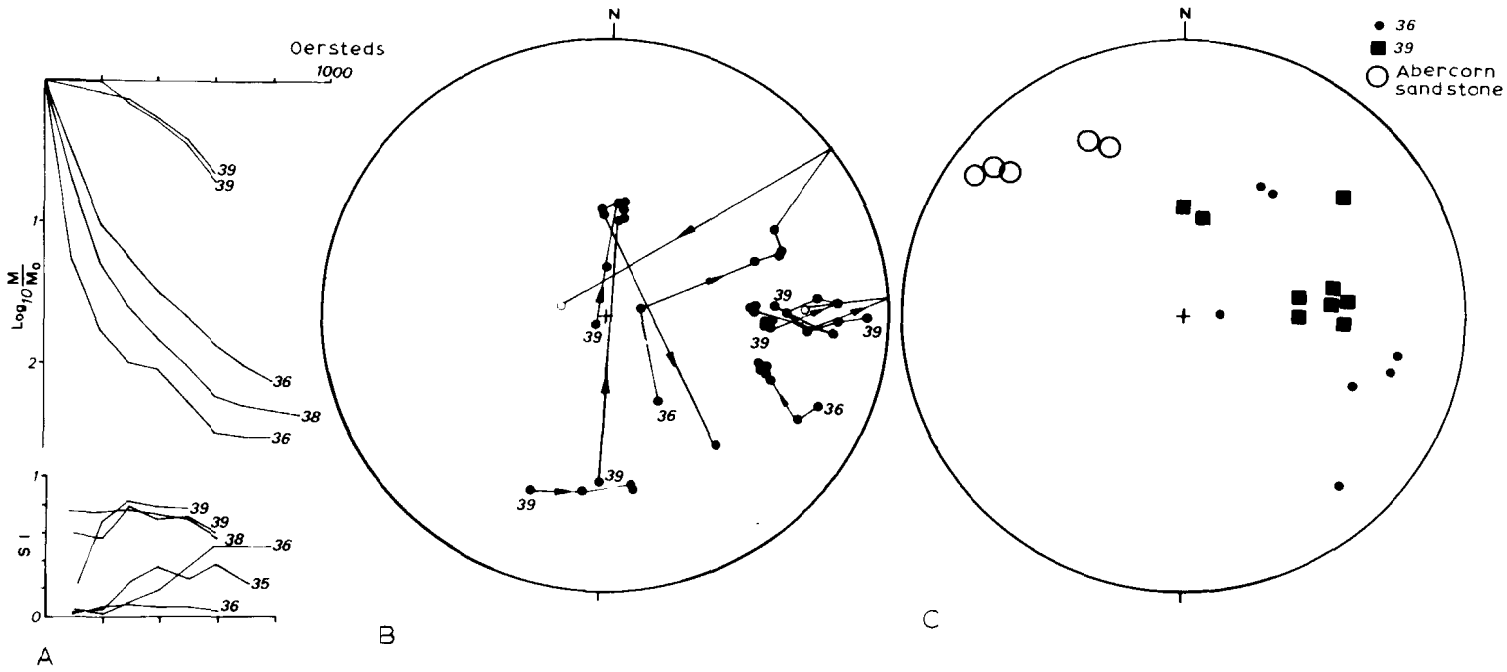


Fig. 5. A. Behaviour of magnetic moments and stability indices of samples of Mbala dolerites with a.f. demagnetization. B. Directional behaviour of pilot samples with progressive a.f. demagnetization in steps of 100 Oe. C. Cleaned sample mean directions from sites 36 and 39 in Mbala dolerites and total NRM directions of five samples from the Abercorn Sandstone.

combine to give a mean direction of $D = 84°$, $I = 40°$ ($k = 19.3$). These two sites have the lowest Qo-ratios of the whole collection (0.5—16, with the majority about 1, and comparing with values mostly in the range 2—23 for the remainder of the collection), suggesting that the insignificant sites have large secondary moments, probably acquired by lightening strikes and which the demagnetization treatment has not been successful in removing.

Ikorongo group, Serengeti, Tanzania

This group crops out sporadically in the area between the Serengeti region and Lake Victoria in Tanzania (Fig. 1) where it overlies metamorphic rocks belonging to the Nyanzian System, its molasse equivalent the Kavirondian System (> 2200 m.y.), and later intrusive dolerites (Harpum, 1955; Quennell et al., 1956). It includes about 1500 m of sediments (Macfarlane, 1969) comprising a basal quartzite overlain by siltstones and sandstones, many of the latter showing ripple marks and current bedding (Macfarlane, 1969; Stockley, 1936).

Six sites have been sampled in red to purple siltstones near the base of the sequence between 25 and 35 km northeast of Seronera. Directions of total NRMs are shown in Fig. 6A. Pilot samples typically lose up to about 30% of their moments with progressive treatment to 1000 Oe with only slight change in direction of NRM in most cases. Remaining samples have been demagnetized in fields of 800 or 1000 Oe but grouping after this treatment ($k = 5.6$) is slightly poorer than at the total NRM level ($k = 5.8$, Fig. 6B). Group mean direction after treatment is $D = 132°$, $I = 53°$ with a palaeomagnetic pole at 84°E,35°S (Table IV).

Buanji Series, southern Tanzania

The Buanji Series crops out in southern Tanzania (Fig. 1) comprising 2000 m of predominantly fluviatile sediments (Harpum, 1955; Stockley, 1948). Six sites collected between 34.0°E,8.9°S and 34.2°E,8.9°S in the Lower Buanji Series defined by Harpum (1955) give well-grouped total NRM directions (Fig. 6C) with a mean of $D = 356°$, $I = 21°$ ($k = 58$). To examine stability of NRM, samples from four sites have been partially thermally demagnetized to 450°C and two have been a.f. cleaned. Samples were stable to this treatment showing loss of no more than a few percent of their total moments with no significant change in direction. The mean direction of six sites after treatment is $D = 358°$, $I = -20°$ ($k = 60$), and although close to the present geomagnetic field direction in the region, is thought to be primary for three reasons. Firstly, the sediments are part of a thick sequence of fine-grained siltstones and remagnetization by solution seems unlikely; secondly the directions are highly stable to thermal demagnetization although this does not of course completely preclude the possibility of remagnetization, and thirdly other studies which include reversals of magnetization and a fold test

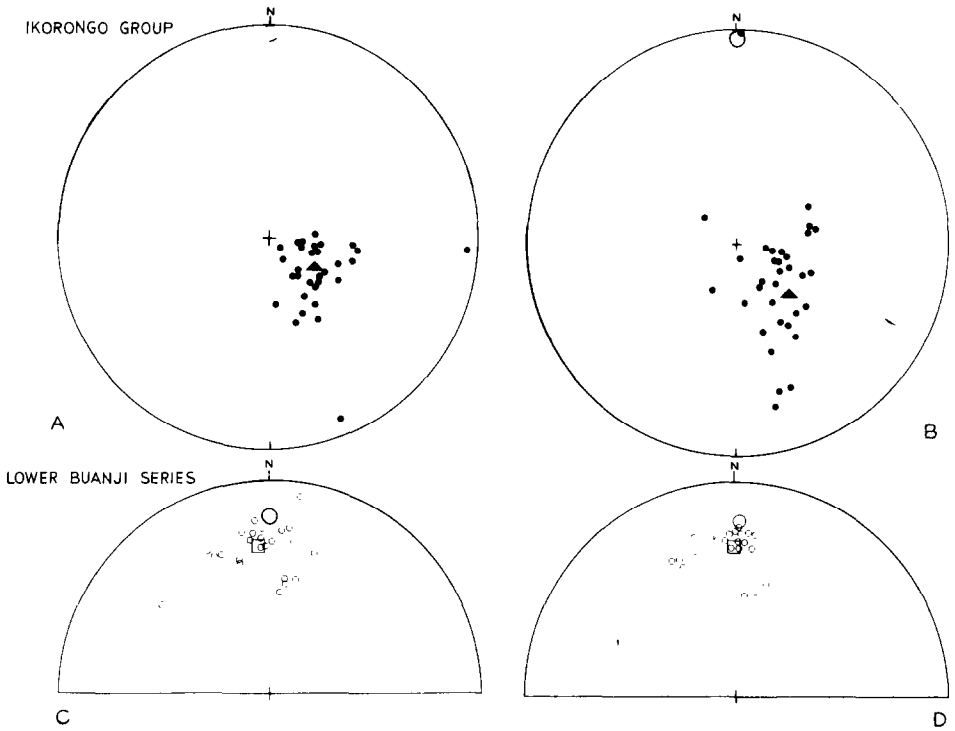


Fig. 6. Distribution of NRM directions from Ikorongo Group sediments. A. Before cleaning. B. After cleaning. C. The Lower Buanji Series before cleaning. D. After cleaning. The squares or triangles are the site group mean directions, and the circles are the directions of the present mean dipole field in the sampling areas.

have shown that the pole occupied a position close to its present one with respect to Africa in part of Late Precambrian and Cambrian times (next section).

INTERPRETATION OF PALAEOMAGNETIC POLES

The apparent polar wander (a.p.w.) curve for Africa during Late Proterozoic time was initially outlined by results from central Africa (Brock and Piper, 1972) and has recently been defined in greater detail (Piper et al., 1973). Certain age inconsistencies in the curve defined by these latter authors for Late Precambrian—Early Palaeozoic time have been reconciled by McElhinny et al. (1974) by envoking a large closed loop between 630 m.y. and Ordovician times; they show that this interpretation is supported by palaeomagnetic data from other Gondwana continents. The a.p.w. curve of Piper et al. (1973) with modifications of McElhinny et al. (1974) is accepted here as a basis for interpretation of pole positions derived from the present study. It is sufficient to note that the findings of these authors strongly suggest that a single a.p.w. curve applied to the bulk of Gondwanaland in Late Proterozoic and Early

TABLE III

Lower Buanji Series, southern Tanzania

Site	Number of samples	D	I	R	k	α_{95}	Cleaning field or temp.	Number of samples	D	I	R	k	α_{95}	Virtual geomagnetic pole position	
														°E	°N
A	4	359	−17	4.0	189	7	450°C	4	359	−18	3.9	191	6	266	89
B	4	353	−20	3.6	7	37	450°C	4	1	−18	4.0	103	9	141	89
C	4	6	−24	3.9	25	18	450°C	4	5	−26	3.9	41	14	168	83
D	3	7	−28	3.0	56	17	450°C	3	5	−25	2.9	34	22	163	83
E	7	338	−22	6.9	42	9	1000 Oe	7	337	−20	6.8	34	10	299	68
F	3	355	−16	3.0	217	8	1000 Oe	3	358	−17	2.9	166	9	311	88
	Combined mean direction (6 sites):						palaeomagnetic pole							palaeomagnetic pole	
	6	356	−21	5.9	58	9	273°E,86°N	6	358	−20	5.9	60	9	263	87
					($d\varphi$ = 4.9, $d\chi$ − 9.4)							($d\varphi$ = 4.9, $d\chi$ = 9.3)			

TABLE IV

Ikorongo Group, Serengeti, northern Tanzania

Site	Number of samples	Total NRM results					Cleaning field (Oe)	Number of samples	Cleaned results					Virtual geomagnetic pole	
		D	I	R	k	α_{95}			D	I	R	k	α_{95}	°E	°N
1	6	102	51	5.3	7	28	1000	7	138	31	5.1	3	41	103	−46
2	5—	99	45	4.5	8	28	1000	5	84	48	4.4	7	32	95	−4
3	6	103	68	5.0	240	4	1000	6	106	67	6.0	243	4	75	−12
4	5	127	72	4.9	62	9	1000	5	137	66	4.9	55	10	67	−31
5	3	165	59	2.9	21	27	800	3	166	46	2.9	38	20	62	−62
6	6	139	51	5.9	54	9	800	5	144	42	5.0	138	7	90	−49
														palaeomagnetic pole	
								6	132	53	5.6	13	19	84	−35
Combined mean direction (6 sites):							palaeomagnetic pole								
	6	121	60	5.8	21	14	80°E,25°S								
			($d\varphi$ = 17.1, $d\chi$ = 22.6)							($d\varphi$ = 18.8, $d\chi$ = 27.2)					

Palaeozoic times; the mobile belts formed during this interval appear to have developed between cratonic regions as ensialic belts without movements of the forelands on a continental scale.

The path of the pole relative to Africa from c. 1000 m.y. to Ordovician times is illustrated in Fig. 7 with the available poles from Africa plotted. The two mineral ages of 940 and 678 m.y. for the Mbala dolerites (Vail et al., 1968) give only a general indication of their true age, but the pole from the dolerites is identical to that obtained from dolerites intruding the Bukoba Formation 700—900 km further north in Tanzania and dated at 806 ± 30 m.y. (Piper, 1972). Agreement of the poles suggests that the Mbala dolerites are part of the same episode of volcanic activity as these dolerites and the related Gagwe Amygdaloidal Lavas of the Bukoban System (813 ± 30 m.y.) 400—600 km to the north. This age for the Mbala dolerites also defines a minimum age for the Abercorn Sandstone Formation, the VGP for which is close to the pole from the Bukoba Sandstone Formation of Tanzania dated at > 995 m.y., and giving additional support to a correlation which has already been made on general geological grounds by a number of workers (e.g. Halligan, 1962); it can therefore also be correlated with post-orogenic rocks of the Irumide Belt.

Cahen and Snelling (1966) give an age of 743 ± 30 m.y. from a biotite in the syenite portion of the Mbozi complex. Ray (1970) has obtained a whole-rock Rb/Sr age of 685 ± 62 m.y. for the related Songwe syenite of northern Malawi, and Cahen and Snelling give a lead age of 655 m.y. for the Ilomba syenite of this region. There seem to be three possibilities from the evidence of the three age dates: either (1) the published age for the Mbozi complex is slightly too old; or (2) there is a real difference in age between the Mbozi and north Malawi complexes; or (3) the ages of the north Malawi bodies have been updated by the "Pan African" events which are also evidenced in this area. In either case the Mbozi pole is probably the earliest record of the movement of the pole from present low latitudes, defined by poles from the upper part of the Bukoban System to present high latitudes, and implies apparent polar movement during part or all of the interval c. 800—740 m.y.

The Plateau Series of sites 1—10 occurs in downfaulted blocks margining the Lake Tanganyika rift, and is younger here than the Abercorn Sandstone. It may postdate the Mbala dolerites since these are not found intruding it. The two pole positions derived from the Plateau Series at Mpulungu in the axis of the Tanganyika rift are almost certainly younger than the pole positions derived from the higher and lower groups of sites from the margin of the rift. When plotted as a function of age, the four pole positions from the Plateau Series crudely define an apparent polar wander path (Fig. 7) the latter two poles of which lie in the vicinity of West Africa close to pole positions already recognized as typical of the latter part of Early Palaeozoic time (McElhinny et al., 1968). This is consistent with discovery of microfossils of Late Cambrian/Early Ordovican age in rocks at the same stratigraphic age as the palaeomagnetic sites at Mpulungu (M. Vamdova, personal communication,

1974). These poles thus postdate those derived from the Ntonya Ring Structure of Malawi (630 ± 24 m.y.; Briden, 1968) and the Late Precambrian Sijarira Group of Rhodesia (Reid, 1968). That however VGPs from the Plateau Series in part predate the similar pole from the Hook intrusives of Zambia (500 ± 17 m.y.; Brock, 1967) is evidenced by Early Cambrian poles from the Fish River Series of South West Africa (Piper, 1974), a Middle Cambrian pole from Morocco (Helsley, 1965) and a pole from the Doornpoort Formation interpreted as dating from the Damaran folding (550—500 m.y.; Piper, 1974), in addition to other Gondwanaland data (McElhinny et al., 1974). Identification of the loop in the a.p.w. curve between c. 630—450 m.y. is also supported by Cambrian to Ordovician poles from South America (Thompson, 1973) which when corrected for Phanerozoic drift define apparent polar movement closely comparable to that identified from African and other results.

Palaeomagnetic evidence is consistent with sedimentation of the Plateau Series having continued to Early Palaeozoic times in the Lake Tanganyika region, and it may thus correlate in part with the Kundelungu System of Katanga and the Luapula Beds of the Zambia—Zaire border. The two significant sites from the southern outcrop however do not agree with VGPs obtained from the Lake Tanganyika outcrop, and the better grouped of these (17) lies near the a.p.w. curve at 900—1000 m.y. It is also close to the pole from the Abercorn Sandstone and indeed the lithology (medium to coarse-grained sandstone) is more like that of the type Abercorn Sandstone than the fine-grained Plateau Series of sites 1—14; however, as a single site only it is not possible to exclude correlation with the 630—450 m.y. loop as an alternative possibility.

Dating of these formations provides a minimum estimate for downthrow at the southern end of the Lake Tanganyika rift. The Abercorn Sandstone Formation is observed at altitudes of 2200 m east of Mbala. Assuming a major gap in sedimentation between c. 900 and 700 m.y. (during which intrusion of the Mbala dolerites took place) and a conservative estimate of 200 m of deposition in the interval Late Precambrian—Early Palaeozoic, downthrow of the rift in the vicinity of Mpulungu amounts to 1500 m. Since the Abercorn Sandstone Formation crops out continuously across the high ground at the southern end of the rift with no major fault displacement, this downthrow may be accommodated by conjugate normal faults comparable to those mapped to the northwest (Fig. 2A).

The Ikorongo Group pole lies close to the a.p.w. curve between 900 and 1000 m.y. which is therefore taken as a probable age of the Group and confirms that it correlates with the lower part of the Bukoban System; such a correlation has previously been made on lithological grounds (Harpum, 1955; Halligan, 1962). The present outcrop of the Group lies close to the foreland of the Mozambique mobile belt (Hepworth, 1972) and Macfarlane (1969) correlates formations of the Group with the lithologically similar sequences of metasediments comprising the Soit Ayai Group involved in the Mozambique

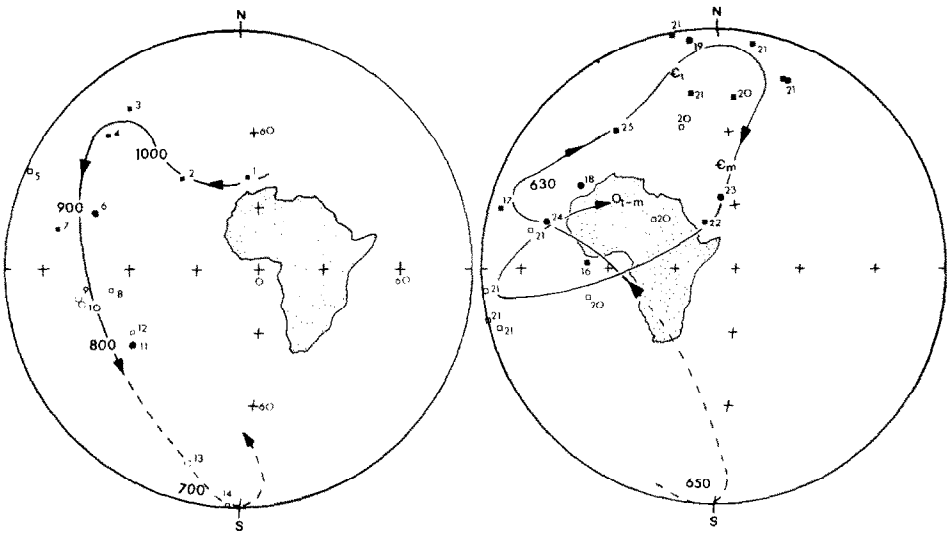


Fig. 7. Apparent polar wander path for Africa from c. 1000 m.y. to Ordovician time based on Piper et al. (1973) and McElhinny et al. (1974). Poles from Africa derived from igneous units are indicated by circles and from sedimentary units by squares; open symbols are poles and closed symbols are antipoles. The poles are: *1*, Auborous Formation, South West Africa (< 1020 m.y.; Piper, 1974); *2*, Bukoba Sandstone Formation, Tanzania (> 995 m.y.; Piper, 1972); *3*, Abercorn Sandstone, Zambia; *4*, Plateau Series (1), Zambia; *5*, Ikorongo Group, Tanzania; *6*, Dykes, Klein Karas, South West Africa, (Piper, 1974; 878 ± 41 m.y.); *7*, Kigonero Flags, Tanzania (Piper, 1972; > 890 m.y.); *8*, Malagarasi Sandstone, Tanzania (Piper, 1972); *9*, Mbala Dolerites, Zambia; *10*, Bukoban dolerites, Tanzania (Piper, 1972; 806 ± 30 m.y.); *11*, Gagwe Amygdaloidal Lavas Tanzania (Piper, 1972; 813 ± 30 m.y.); *12*, Manyovu Red Beds, Tanzania (< 813 ± 30 m.y.); *13*, Mbozi complex, Tanzania (744 ± 30 m.y.); *14*, Lower Buanji Series, Tanzania; *15*, Pre-Nama dykes, South Africa (653 ± 70 m.y.; Piper, 1974); *16*, Sijarira Group, Rhodesia (Late Precambrian; Reid, 1968); *17*, Klipheuvel Formation, South Africa (Lower Cambrian; (?)); *18*, Ntonya ring structure, Malawi (630 ± 24 m.y.; Briden, 1968); *19*, Sabaloka ring structure, Sudan (540 m.y.; Briden, 1973); *20*, Plateau Series, Zambia; *21*, poles from individual sites of the Fish River Formation, South West Africa (Piper, 1974), this formation is of probable Early Cambrian age and sites do not have a Fisherian distribution, possibly because they correlate with different parts of the Late Precambrian to Cambrian shift; *22*, Doornpoort Formation, South West Africa (Piper, 1974; pole dates from interval 550—500 m.y.); *23*, Middle Cambrian lavas, Morocco; *24*, Hook intrusives, Zambia (500 m.y.; Brock, 1967); *25*, Table Mountain Formation (Ordovician).

folding 40—50 km to the east. This result may thus give some indication of the age of the sediments involved in this part of the Mozambique belt; it also provides a minimum age for extensive dolerites which intrude the basement rocks of the area (Stockley, 1936). The Kisii Series of western Kenya lies in close proximity (Fig. 1) to the Ikorongo Group but is lithologically distinct and includes a large volcanic component. The VGP obtained from the volcanics (Brock et al., 1972) falls on the African a.p.w. curve at c. 1200 m.y. and remote from poles derived from the Bukoban System with which it has

hitherto been correlated. Renewed attempts to date the Kisii volcanics using the $^{39}Ar/^{40}Ar$ technique also suggest an age of about 1200 m.y. (Charlton, 1973).

The pole from the Buanji Series apparently falls on the a.p.w. curve between the pole from the Mbozi complex and the pole from pre-Nama dykes of South Africa dated at 653 ± 70 m.y. (Fig. 7); it may alternatively correlate with an extremity of the loop recognized in the a.p.w. curve between 630 and 450 m.y. giving the Series an inferred Early Cambrian age although against this is the negative evidence that the Bunaji Series has not so far been reported as fossiliferous. In either case the Buanji Series proves to be younger than the type Bukoban System and such a correlation can no longer be substantiated. It has been affected by two folding episodes, firstly from the ESE and latterly from the WSE (Harpum, 1955), and sedimentation terminated with extensive intrusion of gabbro and extrusion of basalt. In view of the Late Precambrian or Cambrian age of the Series these episodes may then correlate with tectonic episodes of the Lufilian—Mozambique mobile belts bordering the region (Fig. 1).

CORRELATION OF PRECAMBRIAN FORMATIONS AND RELATION TO TECTONIC EPISODES

The interpretations placed on the pole positions of this study in the preceding section assume that the directions of magnetization isolated from the sediments date from deposition or lithification of the rocks concerned. They will not be valid if magnetizations are secondary and acquired in later times. Unfortunately many of the platform sediments have not been deformed sufficiently for effective application of a palaeomagnetic fold test, although presence of a reversal of magnetization in one case argues for syngenetic magnetization. The bulk of the red sediments sampled for this study are siltstone-grade or finer and there is no realistic case for remagnetization by solution since lithification; the reverse process of reduction is strictly confined to joint planes and weathered profiles. Exceptions to this rule are sandstone samples which are either porous or have been cemented since lithification. Of the present collection sites 16, 17 and 45 fall into this category; in these cases the possibility of remagnetization should be considered, and conclusions drawn from directions of magnetization regarded as tentative.

With this proviso stated, the palaeomagnetic data can be combined with radiometric and geologic information to give a correlation scheme, and this is illustrated diagrammatically in Fig. 8. In this diagram the known or inferred ages of the basement rocks on which the platform sediments lie are also indicated.

Several conclusions can be drawn from these correlations. Firstly, rather uniform conditions of sedimentation appear to have existed over this part of Africa for a period of the order of 500 m.y. These formations are predominantly fluviatile or littoral arenaceous and argillaceous sediments; the palaeomag-

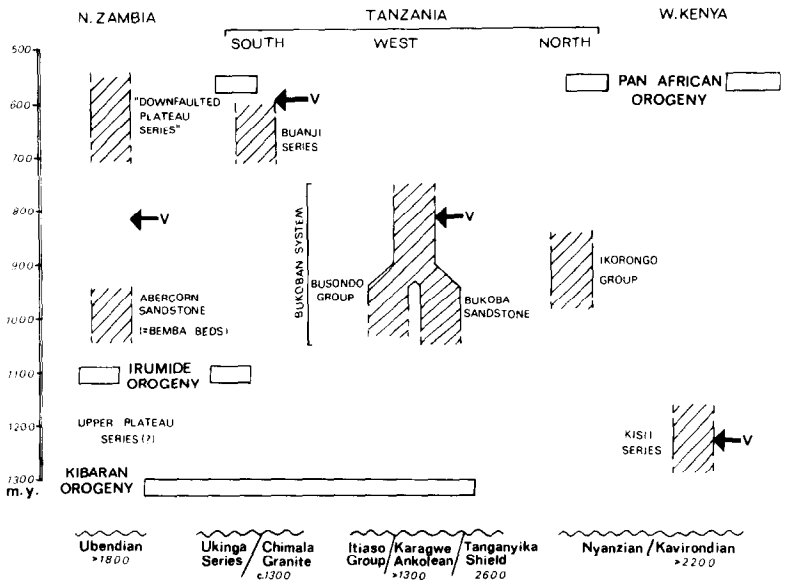


Fig. 8. The correlation of Precambrian platform sediments of east-central Africa based on palaeomagnetic and radiometric data. The *V*'s indicate volcanic episodes, and the horizontal bars are purely diagrammatic representations of the regional influence of orogenic episodes. In particular the Pan African orogeny had a wider age range than shown here and is represented by tectonic events in central Africa before 700 m.y.; its effects in the type of these platform sediments was, however, small and confined largely to a folding of an Ikorongo Group and the Buanji Series. The ages of the basement on which the platform sediments lie are indicated at the bottom of the diagram.

netic data show that they were all deposited in low latitutes and red sediments are ubiquitous, although limestones are thin and uncommon.

Secondly, these formations are mostly too late to be related to preceding orogenic episodes as molasse deposits, and indeed their characteristics are generally not commensurate with such an explanation. They contrast markedly with geosynclinal formations involved in the Kibaran—Irumide folding (1300—1100 m.y.) whose minimum ages are still currently only defined by intrusive igneous rocks. The line of the Kibaran belt in western Tanzania, Rwanda, Burundi and Uganda was the site of thick miogeosynclinal sedimentation prior to the tight folding of this belt at c. 1300, and the type Karagwe-Ankolean of Tanzania (Stockley and Williams, 1938) includes over 13,000 m of orthoquartzites and greywackes and may reach 20,000 m in Uganda. The Itiaso Group lying below the Bukoban system in west-central Tanzania and correlated with the Karagwe-Ankolean (Teale, 1931) exceeds 3000 m. Greywacke-type sedimentation had, however, largely terminated by the time the Bukoban System was deposited and only a few greywackes occur in the basal Masontwa Group (Halligan, 1962). In southern Tanzania the Ukinga Group (> 1300 m.y.) lying below the Buanji Series comprises over 3000 m of mio-

geosynclinal sediments (Harpum, 1955). Other major formations which may correlate with this Group are the Mafingi Series of northern Malawi (> 1200 m.y.; Fitches, 1968) and the Konse Group of southern Tanzania both of which appear to be involved in the Irumide folding (Vail et al., 1968).

Thirdly, deposition of these formations was a consequence of intermittent uplift of the Tanganyika craton between about 1000 m.y. and Cambrian times. Several of these formations show large-scale cyclic repetition of lithology. At least eight regressive cyclothems (shales to sandstones) are present in the Bukoba Sandstone Formation, and the three units of its lateral equivalent in the south of the Bukoban outcrop (Uruwira Sandstone—Nyanza Shale—Malagarasi Sandstone; Halligan, 1962), form a comparable single repetition of lithology. The Upper Plateau Series also shows a similar large-scale repetition (Page, 1960). The Bukoban System was deposited in an intermittently subsiding basin on heavily eroded older rocks, and four unconformities and one major disconformity occur within the c. 250 m.y. represented by deposition of the System. The former separate episodes of concentric folding the axes of which trend from NNE in the north of the outcrop (Macdonald, 1963) to NNW in the central part and to NW in the southern part (Halligan, 1962). They thus parallel the axis of deposition of the System and the margin of the Tanganyika craton; also the folds are typically asymmetrical with steeper limbs always on the side furthest from the craton. The position of the subsiding basin around the margin of the craton was also a line of major basaltic volcanism c. 810 m.y. (Bukoban dolerites—Gagwe Amygdaloidal Lavas—Mbala dolerites).

Unfortunately little information is currently available on the source of formations under discussion here except that the Abercorn Sandstone is known to have been derived from the north and east (Page, 1960). However geological setting and distribution of the formations suggests that the Tanganyika craton was a region of relative uplift during the period of their formation, and clear evidence for this is seen in numbers of the Busondo Group in the lower part of the Bukoban System which thicken away from the craton (Halligan, 1962). Within the area of the craton the only platform-type sediments are found in a small downfaulted outcrop of Kingongolero (34.8° E, 6.4° S); where an age of 1365 ± 60 m.y. on intruding dolerite (Cahen and Snelling, 1966) suggests that they are considerably older than formations preserved around the margin of the craton.

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